A systematic review of construction cost estimation in housing: method, system requirement and variables

1.0 Introduction

The housing sector remains one of the most cost-sensitive segments of the built environment (Ma et al, 2020). Accurate cost estimation is essential for project feasibility, affordability, and timely delivery (Amade & Akpan, 2014). In the housing sector, persistent housing deficits, rapid urbanisation, inflationary pressures, and volatile construction input prices have intensified the consequences of inaccurate cost forecasts (Udensi et al, 2024). Research findings (Ahiaga-Dagbui, D. D. 2014; Eliasson, 2025) reports significant cost overruns, delayed completions, and abandoned developments, largely attributed to poor early-stage cost estimation, inadequate data, and reliance on manual, experience-based estimating practices (Asiedu & Adaku, 2020). These challenges underscore the urgent need for more reliable, automated, and data-driven cost estimation approaches tailored to the housing context.

Construction cost estimation has traditionally relied on methods such as approximate estimating, elemental cost planning, and bills of quantities (Mislick & Nussbaum, 2015). These approaches remain central to professional practice; however, their effectiveness is highly dependent on the availability of detailed design information and stable market conditions (Dandan et al, 2020). Housing projects especially in developing countries, are frequently initiated with incomplete designs and exposed to fluctuating material prices, exchange rate instability, and informal labour markets. Conventional estimating methods often fail to capture these real-life cost dynamics (Azahar & Abdul Karim, 2023).

In recent times, advances in automation and digital technologies have significantly reshaped construction cost estimation research globally (Wang et al, 2020). Construction cost estimation literature documents increasing use of statistical modelling, machine learning, and artificial intelligence techniques such as artificial neural networks, support vector machines, genetic algorithms, and ensemble models to improve cost prediction accuracy in other building categories, including; healthcare, commercial and educational facilities (Cao et al, 2020; Habib et al., 2025). These approaches are particularly effective in handling non-linear relationships between cost drivers, uncertainty, and incomplete information, making them suitable for early-stage cost estimation (Ofori-Boadu, 2015).

Building Information Modelling (BIM) has further transformed automated cost estimation by enabling integrated quantity take-off and enhanced collaboration between design and cost information (Safaa-Eldin et al, 2024). Research have reported that BIM-based cost estimation improves transparency, reduces human error, and supports scenario-based decision-making (Gholamzadehmir et al, 2020).

Housing is a unique building category due to its diversity in design, scale, and occupant requirements (Bodendorf et al, 2021). Unlike commercial or institutional buildings, the methods, variables, and data necessary for developing accurate housing cost estimating models remain unclear (Eliasson, 2025). This uncertainty presents a significant challenge for researchers and practitioners seeking to create reliable, context-specific models that can capture the complexity of housing construction costs.

This study addresses this gap by conducting a systematic literature review (SLR) of construction cost estimation research, with a particular emphasis on automation-driven approaches and their suitability for housing cost estimation. Adopting established systematic review protocols, the study synthesises peer-reviewed articles published in leading journals, to examine construction cost estimation methods, data

requirements, automation levels, variables and reported performance outcomes with a view to outline methods, data requirements and variables suitable for housing construction cost estimation. The review traces the evolution from traditional estimating practices to BIM-enabled with the exclusion of machine learning, and artificial intelligence-based models, highlighting trends, limitations, and contextual challenges specific to housing.

Presenting a structured synthesis of existing research, this study contributes to both academic and industry practice. For researchers, it identifies methodological gaps, data challenges, and underexplored opportunities for applying automation in housing cost estimation. For practitioners and policymakers, the findings offer evidence-based insights into how automated cost estimation systems can improve cost predictability, support housing affordability strategies, and enhance decision-making in housing construction budgeting. This review aligns with the core objectives of scopus indexed by advancing knowledge on how digitalization and intelligent systems can address real-world construction cost challenges in developing economies.

RQ1: what are the methodologies for cost estimation in housing?

RQ2: what are the variables for cost estimation in housing?

RQ3: what are the system requirements for cost estimation model in housing?

2.0 Research Method

This study used a qualitative research approach and applied a systematic literature review (SLR) to identify the methodology, system requirement and variables for construction cost estimation in housing. A systematic literature review is a structured and transparent method that allows existing research on a topic to be identified, assessed and synthesised in a consistent way (Templier & Pare, 2018). Research conducted over time, SLR is suitable for defining the current boundaries of knowledge within a specific field.

According to Marzi *et al*. (2025), SLR helps to build a strong foundation for theory development, summarises areas where substantial research already exists, and highlights topics that require further investigation. SLR has been widely applied in architectural engineering and construction (AEC) research to review and integrate published studies (Shinde et al., 2023). The SLR in this study was carried out using a structured process that included a systematic search of literature, screening of relevant studies, critical evaluation of selected papers, extraction of key metadata, and detailed content analysis, as displayed in Figure 1.

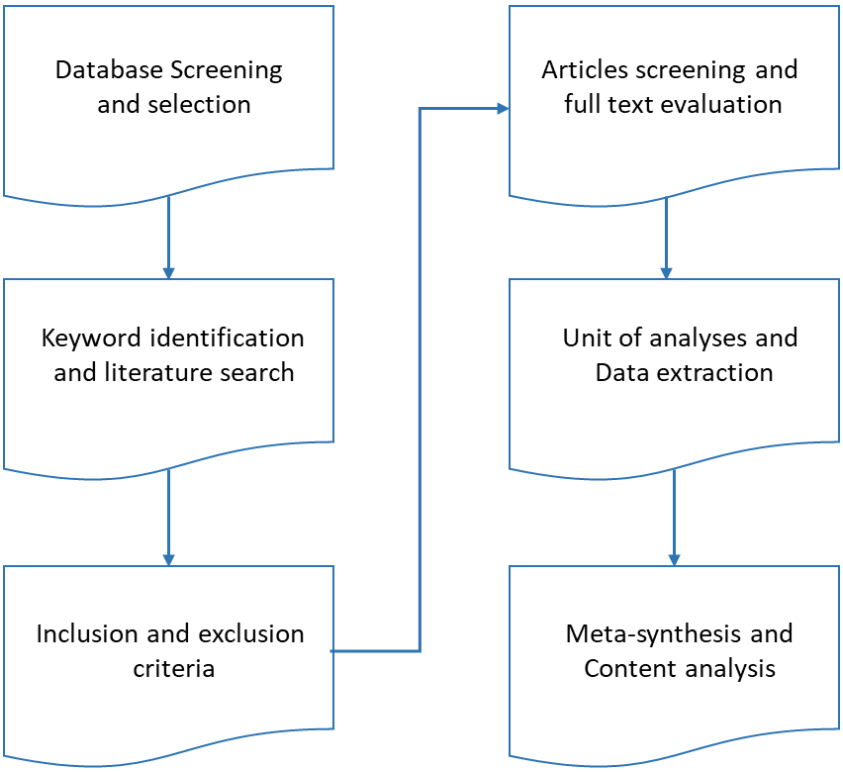

Figure 1. Research design

Source: Wuni *et al.* (2019)

2.1 Systematic literature search

Several well-established AEC databases and academic libraries were reviewed to determine which offered the most comprehensive, accurate and relevant coverage. The keywords was identified from the research objectives, Boolean operators "AND" , "OR" , "NOT" was used to connect the keywords, Using an identical set of keywords, initial searches were undertaken across OpenAlex, Elsevier's Scopus, Clarivate Analytics' Web of Science, the ASCE Library, Engineering Village, Taylor & Francis, and Emerald Insight.

The initial results showed that many of the identified studies appeared repeatedly across the different databases. However, OpenAlex provided the widest coverage of relevant publications. In addition, OpenAlex was found to be easy to use, allowed flexible filtering of search results, and offered useful features such as citation information and reference linking. For these reasons, OpenAlex was selected as the sole database for the literature search. OpenAlex is an open scholarly database that gathers and organises research metadata by automatically harvesting DOI-based records mainly from Crossref and other sources Scopus included. This decision is consistent with SLR in the AEC field that also relied on OpenAlex for sourcing articles (Couto & Baltazar, 2025).

After selecting database, commonly used alternative terms for "Cost estimation", "Infrastructure", "variables" and "Artificial Intelligence" were identified from existing studies. The keyword list was refined and adjusted continuously throughout the review process. The final and most suitable combination of keywords was confirmed at the end of the review. The complete OpenAlex search string is presented below.

(TITLE-ABS-KEY ("Cost estimation" ***OR*** *"Cost"* ***OR*** *"Cost estimates"* ***OR*** *"Cost estimating systems"* ***OR*** *"Cost estimating models"* ***OR*** *"Database"* ***OR*** *"Cost data management"* ***AND*** *"Infrastructure"* ***OR*** *"Housing"* ***OR*** *"National housing"* ***OR*** *"Housing models" "Residential"* ***OR*** *"Construction"* ***OR*** *"Building"* ***AND*** *"Variables"* ***OR*** *"Macro-economic indices"* ***OR*** *"Equipment"* ***OR*** *"Plant"* ***OR*** *"Location"* ***OR*** *"Inflation"* ***OR*** *"Materials"* ***OR*** *"labor"* ***OR*** *"Labour"* ***OR*** *"Markup"* ***OR*** *"Land" "Preliminaries"* ***NOT*** *"Artificial intelligence"* ***OR*** *"Machine learning"* ***OR*** *"Deep learning"* ***OR*** *"neural networks")).*

As shown in the search string, several limits were applied. Only journal articles and articles-in-press were included, the search was restricted to publications written in English, and only journal sources were considered. These criteria were applied to ensure consistency. No time limits were placed on publication year. Using this search strategy, 985 records were initially retrieved from OpenAlex. To ensure that recently published studies were not missed, the search was repeated shortly before submission of the manuscript.

2.2 Inclusion and exclusion criteria

Nightingale (2009) notes that a systematic literature review must clearly state its inclusion and exclusion criteria so that the study can be checked and repeated by other researchers. This study defined clear criteria to screen and filter the 985 records retrieved from the database and to identify the final sample.

An article was included if it met two conditions: first, it had to be an empirical study that examined construction cost estimation in housing or building; and second, it had to be published in a peer-reviewed journal. Conference papers were excluded with the exception of one paper that was deem fit to be included., as they are often considered to undergo less rigorous peer-review.

Using these criteria, the titles and abstracts of all 985 records were first reviewed to identify potentially relevant studies. This initial screening resulted in 228 articles being selected for further assessment. A full-text review was then carried out, which led to the inclusion of 27 articles. The screening process is illustrated in Figure 2. Although the number of selected articles appeared relatively small, it compares well with AEC reviews that used samples of 16 studies (Newaz *et al*., 2018) and 27 studies (Osei-Kyei & Chan, 2015).

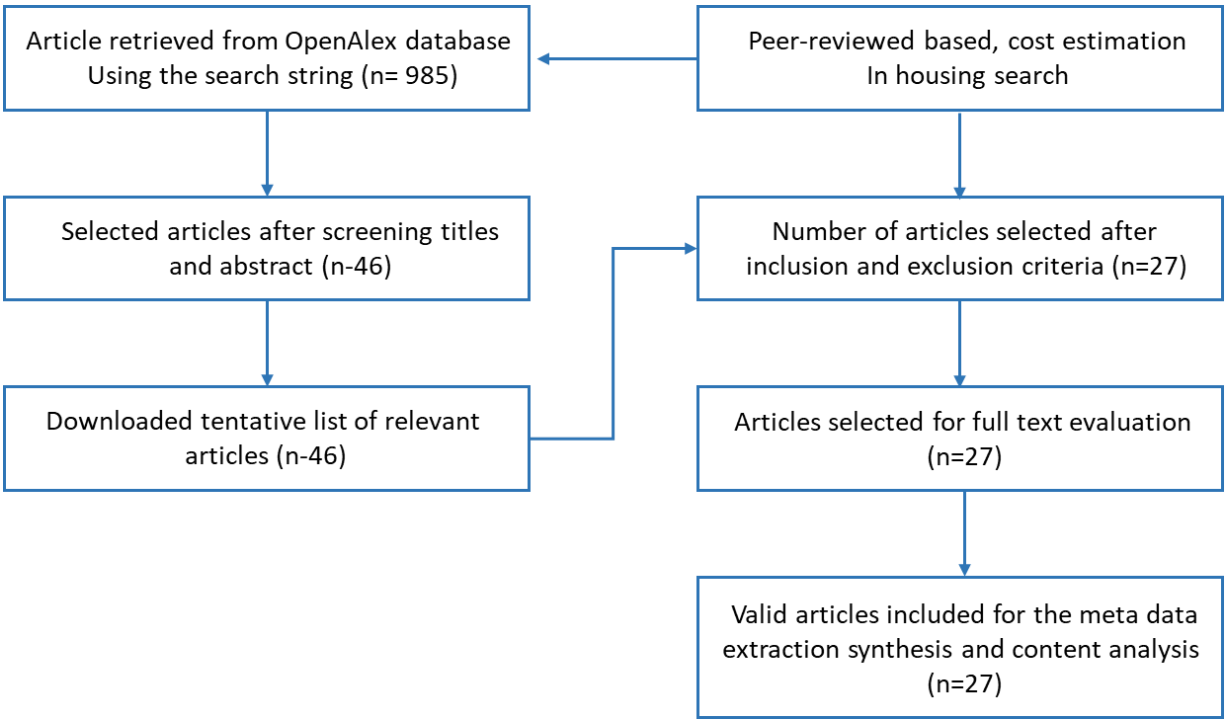

Figure 2. Research method flow chart

27 articles were included in the final review. This sample size was considered sufficient, given that cost estimation in housing excluding artificial intelligence research is still developing in many countries. The full list of the 27 included studies and their identification numbers is presented in Table 1 and referenced throughout the paper.

Table 1. Valid articles selected for analysis

S/No.	Citation	S/No.	Citation
1	Jrade & Alkass (2007)	15	Ji et al. (2019)
2	Sharma et al. (2013)	16	Meharie et al. (2019)
3	Kim et al. (2014)	17	Hyung et al. (2020)
4	Yildiz et al. (2014)	18	Ahn et al. (2020)
5	Zima (2015)	19	Akanbi & Zhang (2020)
6	Liu et al. (2016)	20	Fazeli et al. (2020)
7	Ma et al. (2016)	21	Pujitha & Venkatesh (2020)
8	Tae & Lee (2016)	22	Akanbi & Zhang (2021)
9	Abanda et al. (2017)	23	Le et al. (2021)
10	Akanbi & Zhang (2017)	24	Xiao et al. (2023)
11	Salari et al. (2017)	25	Jassim et al. (2025)
12	Swei et al. (2017)	26	Putranto & Ma'soem (2025)
13	Chau et al. (2018)	27	Solla et al.2025
14	Dang & Le-Hoai (2018).		

2.3 Thematic content analysis

Supporting the extraction and organisation of metadata, this study used meta-synthesis as the main method for structuring the reviewed literature. Unlike the traditional meta-analysis, which focus only on quantitative evidence. Meta-synthesis allows findings from both qualitative and quantitative studies to be combined within a single systematic review (Baker, 2016). Clear unit of analysis were defined to guide the extraction of information from each study. Based on the research questions, key details were collected from each article, including year of publication, journal outlet, methodology, system requirement and variables reported. This information was compiled into a summary table, described as a thematic content matrix with defined units of analysis (Neuendorf, 2018).

The identified methods, system requirement and variables were then grouped and integrated using thematic content analysis. According to Stone (2020), thematic content analysis offers a systematic way of identifying patterns and recurring themes across large volumes of text. Thematic content analysis method helped to align and standardise differences in how cost estimation in housing methods, system requirement and variables were described across the reviewed papers. Similar approaches have been applied in AEC review studies (Newaz *et al.,* 2018; Wuni et al., 2019).

3.0 Findings and discussion

3.1 Cost estimation in housing methodologies

Table 2 shows phase-dependent distribution of cost estimation methodologies. This aligns with construction cost management literature (Wouters et al., 2016; Farouk & Rahman, 2025). Approximate estimate methods are confined to the pre-design stage, reflecting their reliance on limited project information such as floor area and building form. This is consistent with quantity surveying practice, which positions this method as strategic tools for feasibility assessment rather than precise cost prediction (Goodall et al., 2014). The relatively low frequency, suggest limitations in addressing design complexity and uncertainty as projects progress.

Elemental detailed estimate, and BIM-enabled methods dominate the design phase, reinforcing the long-standing principle that cost accuracy improves with increasing design definition (Salk et. Al., 2018). The prominence of ontology-based, database-driven, and BIM-supported estimating approaches presented a digital evolution of traditional bill-of-quantities and elemental cost planning rather than a fundamental methodological shift (Olatunji & Sher, 2014). In practice, these approaches remain central to tendering and cost control, particularly in housing projects where standardisation and repetition support reliable unit-rate development.

Statistical and parametric methods emerge as the most prevalent across both pre-design and design stages, reflecting a growing research emphasis on data-driven decision support under uncertainty. However, their dominance in academic literature is met with a cautious industry adoption, due to issues of data availability and model transparency. Life cycle costing is underrepresented.

Table 2. Phase-dependent distribution of cost estimation methodologies

Methodology	Project phase	Source	Frequency
Approximate estimate	Pre design	[14,21,25]	3

Methodology	Project phase	Source	Frequency
Elemental detailed estimate	Design	[5,6,7,8 15,19,22,23]	8
Statistical & Parametric	Pre-design & design	[2,3,4,11,12,13,16,17,18,24,26]	11
BIM-Based	Pre-design & design	[9,10,20,27]	4
Life Cycle Costing	Strategic Decision	[1]	1

The phase-dependent distribution of cost estimation methodologies, created a need for further development of pre-design approximate cost estimation methods. Approximate estimate are largely confined to the pre-design stage, where critical strategic decisions on project feasibility, affordability, and scope, are made under information constraints. Despite the high frequency and acknowledged limitations in handling design complexity and uncertainty, the approximate estimate remain indispensable because no alternative approach can operate effectively with such minimal inputs.

3.2 Cost estimation in housing variables

Table 3 presents a mapping of cost estimation methodologies to project phases and their associated variables. Approximate estimating methods are confined to the pre-design stage and rely on functional variables such as gross floor area, number of storeys, building height, and location index. This aligns with quantity surveying literature, which emphasizes that early estimates are strategic, supporting feasibility and affordability decisions when design information is limited (Wouters et al., 2016; Farouk & Rahman, 2025). Elemental detailed estimating is dominant at the design stage, where sufficient information exists to break the building work breakdown structure into elements such as substructure, superstructure, external works. The variables identified in Table 3 are consistent with the BESMM4R, RICS New Rules of Measurement and traditional bill of quantities practice, where cost certainty improves as design definition increases. The addition of labour productivity and contractor mark-up reflects tendering conditions, reinforcing the role of elemental estimates as the backbone of cost planning, tender pricing, and cost control in housing projects.

Statistical and parametric methods span both pre-design and design phases, using variables that combine functional attributes with market-related factors. The literature widely supports this hybrid variable structure, as predictive models depend not only on physical characteristics but also on procurement strategy, contract type, market conditions, and risk exposure (Martínez et al., 2020). These methods are increasingly used to complement traditional estimating by improving early-stage forecasting and supporting decision-making under uncertainty, although their practical adoption remains moderated by data availability and transparency concerns. BIM-based and life cycle costing approaches extend the variable set further by incorporating digital and time-dependent dimensions of cost. BIM-based methods rely on object-level quantities, model fidelity, and standardized classification systems to automate and dynamically update cost information. Consistent with 5D BIM literature. Life cycle costing introduced operational, maintenance, and replacement variables to support strategic decision-making over the asset's lifespan, as advocated in ISO 15686-5 (Safaa-Eldin et al., 2024). The limited application of life cycle variables in practice, particularly in housing, reflects a persistent gap between long-term objectives and short-term capital cost priorities highlighted in literature.

Table 3. project-phase based cost estimation methodologies and associated variables

Methodology	Project Phase	Variables
Approximate Estimate	Pre-design	Gross floor area, number of storeys, building height/average floor height, building type, location index, construction duration, quality/standard level, basic structural system, inflation index
Elemental Detailed Estimate	Design	Substructure cost, superstructure cost such as; walls, roof, finishes, doors and windows, mechanical services, electrical services, external works, elemental quantities, unit rates, material specifications, labour productivity, mark-up
Statistical & Parametric	Pre-design & Design	Gross floor area, number of dwelling units, number of floors, building height, project complexity index, location factor, procurement method, contract type, construction duration, historical project costs, market conditions, risk factors, inflation/escalation rates.
BIM-Based	Pre-design & Design	BIM object quantities (walls, slabs, beams, columns), component dimensions, material properties, model Level of Development (LOD), work classification codes (UniFormat/MasterFormat), linked cost database values, design alternatives, design change frequency
Life Cycle Costing	Strategic decision	Initial construction cost, operation cost, maintenance cost, replacement cost, energy cost, service life of components, discount rate, inflation rate; residual value, analysis period.

3.3 Cost estimation model in housing system requirements

Table 4 shows the system requirements reflecting a phase-responsive cost estimation outline that aligns with construction cost management practice. At the pre-design stage, the system prioritises rapid feasibility assessment through functional inputs, acknowledging the limited availability of design information and the strategic nature of early cost decisions. As projects progress into the design phase, the system shifts toward elemental and detailed cost breakdowns, ensuring traceability, transparency, and compliance with standard measurement rules that underpin tendering and cost control in practice. The inclusion of statistical, parametric, and BIM-based capabilities enables the system to enhance traditional estimating by supporting data-driven forecasting, automated quantity extraction, and dynamic cost updates in response to design changes. The life cycle costing module extends the system beyond capital cost estimation to support strategic, long-term decision-making by incorporating operational and maintenance considerations. These requirements emphasise scalability, and interoperability, ensuring the system can accommodate varying information levels, improve early-stage cost reliability, and support informed decision-making across the entire project lifecycle.

Table 4. System requirements for cost estimation across project phases

Methodology	Project Phase	System Requirements
Approximate Estimate	Pre-design	The system shall allow input of high-level project attributes including gross floor area, number of storeys, building height, housing typology, and location index. It shall support indicative construction duration, quality/standard selection, and basic structural system options. The system shall automatically apply inflation indices and generate rapid feasibility-level cost outputs with minimal data dependency.
Elemental Detailed Estimate	Design	The system shall support elemental cost breakdown in accordance with standard measurement rules, enabling input and calculation of substructure, superstructure, finishes, services, and external works. It shall manage elemental quantities, unit rates, material specifications, labour productivity, and contractor mark-up. The system shall provide traceable cost summaries suitable for cost planning, tender preparation, and cost control.
Statistical & Parametric	Pre-design & Design	The system shall incorporate a data-driven estimation engine capable of processing functional, contextual, and historical variables such as building size, number of units, project complexity, procurement route, contract type, market conditions, and risk factors. It shall support regression or AI-based models, enable scenario analysis, and adjust estimates using escalation and inflation parameters.
BIM-Based	Pre-design & Design	The system shall integrate with BIM models to automatically extract object-level quantities, component dimensions, and material properties. It shall recognise model Level of Development (LOD), apply standard classification systems (UniFormat/MasterFormat), and link extracted quantities to an updatable cost database. The system shall dynamically update cost outputs in response to design alternatives and model changes.
Life Cycle Costing	Strategic decision	The system shall support long-term cost analysis by incorporating initial construction costs alongside operation, maintenance, replacement, and energy costs. It shall allow definition of service life, analysis period, discount rate, inflation rate, and residual value. The system shall compute whole-life cost outputs to support strategic and sustainability-oriented decision-making.

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